

Random lifts of K_5 are asymptotically almost surely 3-chromatic

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Machine-generated note

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Abstract

We prove that a uniform random n -lift of K_5 has chromatic number 3 with probability tending to one. A Bregman permanent bound keeps the matching constraints in the second moment of fibre-balanced 3-colourings, making the uniform overlap the unique global maximum; the resulting variance is then exhausted by short cycles, so small-subgraph conditioning yields a colouring asymptotically almost surely.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

An n -lift of a graph G replaces every base vertex by a fibre of n vertices and every base edge by a perfect matching between the corresponding fibres. In a uniform random lift these matchings are chosen independently and uniformly. The OpenProblemGarden entry [1], following Amit, Linial and Matoušek [2], asks verbatim:

Question 1.1. Is the chromatic number of a random lift of K_5 concentrated on a single value?

The possible asymptotic values are 3 and 4. Farzad and Theis proved the analogous 3-colourability result for $K_5 \setminus e$ [3].

Nir and Pérez-Giménez [4] locate the chromatic number of random lifts of K_{d+1} in a two-value window. For three colours their upper theorem requires $d < \ell_3 = 16 \log 2/3 \approx 3.69678$, while their lower theorem starts at $u_3 = 2 \log 3/(\log 3 - \log 2) \approx 5.41902$. Thus $d = 4$, corresponding to K_5 , lies in the gap. Their second-moment optimization relaxes away matching marginals. The argument here instead uses Bregman's permanent inequality [5], in its entropy form due to Radhakrishnan [6], and remains sharp at the uniform overlap when $d = 4$.

Theorem 1.2 (Chromatic number of a random lift of K_5). *Let H_n be a uniform random n -lift of K_5 . Then*

$$\Pr(\chi(H_n) = 3) \rightarrow 1.$$

Idea of proof. Count colourings that use nearly $n/3$ vertices of each colour in every fibre. An entropy inequality and a permanent bound show that only nearly independent pairs of colourings contribute to the second moment. A local lattice calculation evaluates that contribution exactly. Its excess over the squared first moment equals precisely the contribution of short cycles, by the Ihara–Bass determinant identity. The small-subgraph conditioning theorem then shows that a balanced 3-colouring exists with probability tending to one. A first-moment estimate excludes bipartite lifts. [\[full proof\]](#)

2 Balanced colourings and the two analytic inputs

Choose integers t_1, t_2, t_3 summing to n , each differing from $n/3$ by at most 1, and set $p_a = t_a/n$. Let Z_n count proper 3-colourings in which every fibre has exactly t_a vertices of colour a . Write

$$M_n = \frac{n!}{t_1!t_2!t_3!},$$

and let a_n be the probability that a uniform perfect matching between two fibres with these colour counts has no monochromatic edge. Then

$$\mathbb{E}Z_n = M_n^5 a_n^{10}. \quad (1)$$

For two balanced colourings, the overlap in fibre i is the matrix

$$\rho_i(a, b) = \frac{1}{n} |\{x : \text{the two colours of } x \text{ are } a, b\}|.$$

Its row and column sums are p . Independently in the five fibres,

$$\Pr(\rho_i = \rho) = \frac{\prod_{a=1}^3 (t_a!)^2}{n! \prod_{a,b=1}^3 (n\rho(a, b))!}. \quad (2)$$

If $q_n(\rho, \sigma)$ is the probability that a matching between fibres of overlap types ρ, σ respects both colourings, then

$$\frac{\mathbb{E}Z_n^2}{(\mathbb{E}Z_n)^2} = \mathbb{E}_{\rho_1, \dots, \rho_5} \prod_{ij \in E(K_5)} \frac{q_n(\rho_i, \rho_j)}{a_n^2}. \quad (3)$$

Let U be the 3×3 matrix with all entries $1/9$, and define

$$H(\rho) = - \sum_{a,b} \rho(a, b) \log \rho(a, b), \quad L(\rho) = \sum_{a,b} \rho(a, b) \log \left(\frac{1}{3} + \rho(a, b) \right),$$

and $F = H + 2L$.

Lemma 2.1 (Entropy inequality). *There is $c > 0$ such that every nonnegative 3×3 matrix with row sums $1/3$ satisfies*

$$F(\rho) \leq F(U) - c \|\rho - U\|_F^2, \quad F(U) = 2 \log(4/3).$$

Idea of proof. After rescaling a row by 3, the problem is to maximize $\sum_{j=1}^3 g(x_j)$ on the simplex for $g(x) = 2x \log(1+x) - x \log x$. The function has a single inflection point. Concavity handles three small coordinates, while reducing the one-large-coordinate case to a strictly concave one-variable function. The Hessian at the uniform point is negative definite, giving the quadratic gap. [\[full proof\]](#)

Lemma 2.2 (Finite-type matching asymptotic). *Let A be a symmetric zero-one matrix of order s with constant row sum r , put $P = A/r$ and $u = (1/s, \dots, 1/s)$. Assume that $I - P^2$ is positive definite on $\mathbf{1}^\perp$ and that the vectors $e_j - e_k$ for columns occurring together in a row support generate the zero-sum integer lattice. If $Q_A(n; \alpha, \beta)$ is the probability that a uniform matching with integer type counts $n\alpha, n\beta$ uses only allowed type pairs, then, uniformly for $\|\alpha - u\| + \|\beta - u\| = O(n^{-2/5})$,*

$$\begin{aligned} Q_A(n; \alpha, \beta) &= \left(\frac{r}{s} \right)^n \det_{\mathbf{1}^\perp} (I - P^2)^{-1/2} \\ &\quad \times \exp \left\{ \frac{ns}{2} [\|\beta - u\|^2 - \langle w, (I - P^2)^{-1} w \rangle] \right\} (1 + o(1)), \\ w &= \beta - u - P(\alpha - u). \end{aligned}$$

Idea of proof. Choose permitted right types independently and express the matching probability as a ratio of two lattice point probabilities. Exponential tilting gives the quadratic rate function; uniform Fourier inversion supplies the Gaussian prefactor. The lattice hypothesis rules out nontrivial maximum-modulus characters. [\[full proof\]](#)

3 Moment and cycle calculations

Proposition 3.1 (Second moment). *For the variable Z_n above,*

$$\frac{\mathbb{E} Z_n^2}{(\mathbb{E} Z_n)^2} \longrightarrow R := 25 \left(\frac{16}{15} \right)^{20} \left(\frac{15}{23} \right)^8.$$

Moreover $\mathbb{E} Z_n = \Theta(n^{-5})(4/3)^{5n}$.

Idea of proof. Bregman’s inequality and [Lemma 2.1](#) suppress every overlap farther than $n^{-2/5}$ from independence. On the central window, [Lemma 2.2](#) and Stirling turn the sum into a 20-dimensional Gaussian integral. Its five-fibre matrix is $(19I - 4A(K_5))/15$, with eigenvalues $1/5, 23/15, \dots, 23/15$. [\[full proof\]](#)

Let B be the nonbacktracking matrix of K_5 , indexed by its 20 directed edges, and let $X_{\ell,n}$ count unoriented simple ℓ -cycles in H_n . Put

$$\mu_\ell = \frac{\text{tr}(B^\ell)}{2\ell}, \quad \delta_\ell = 2(-1/2)^\ell. \tag{4}$$

Proposition 3.2 (Short-cycle moments and variance identity). *For every fixed L , $(X_{3,n}, \dots, X_{L,n})$ converges jointly to independent Poisson variables of means $(\mu_3, \dots, \mu_L)$. For every finitely supported sequence of nonnegative integers b_ℓ ,*

$$\frac{\mathbb{E} \left[Z_n \prod_{\ell \geq 3} (X_{\ell,n})^{b_\ell} \right]}{\mathbb{E} Z_n} \longrightarrow \prod_{\ell \geq 3} (\mu_\ell (1 + \delta_\ell))^{b_\ell}.$$

Finally,

$$\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 < \infty, \quad R = \exp \left(\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 \right).$$

Idea of proof. Factorial moments count vertex-disjoint lifted cycles; intersecting configurations lose a factor n^{-1} . Under the colouring-size-biased measure, a cycle gains the factor $1 + \delta_\ell$. The Ihara–Bass determinant identity at $u = 1/4$ turns the sum of $\mu_\ell \delta_\ell^2$ into the same determinant as the Gaussian second moment. [\[full proof\]](#)

4 Remarks

Remark 4.1 (What is new relative to the 2021 bound). The relaxation used in [\[4\]](#) has a nonuniform maximizer when $d = 4$. The permanent bound here retains the perfect-matching constraint and has the uniform overlap as its unique maximizer. The referee numerically optimized the exact maximum-entropy coupling as an independent check and found the same maximizer.

Remark 4.2 (Scope). The argument also gives asymptotic 3-colourability for lifts of any fixed simple 4-regular base; it does not assert chromatic number exactly 3 for bipartite bases. The general Open-ProblemGarden question for K_m remains untouched. The entropy lemma is specific to degree 4: its analogue already fails at degree 5, so this method does not resolve random lifts of K_6 .

Remark 4.3 (Computational audit). The referee verified every displayed constant. In particular, $R = 2.9745540512$ and

$$\sum_{\ell=3}^{60} \mu_\ell \delta_\ell^2 = 1.090094126, \quad \log R = 1.090094129,$$

the residual being truncation. In simulations, the 3-colourable fraction was $211/400$ at lift size 3, $293/300$ at size 12, and $200/200$ at size 20; no bipartite lift occurred in more than 1,400 trials. These computations support the proof but do not replace it.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The proof was found by gpt-6-astra, reasoning effort max, in a single pass started on 2026-09-16, for OpenProblem-Garden catalog id chromatic_number_of_random_lifts_of_complete_graphs, campaign leg attacks_opg. (2) An independent adversarial referee agent (claude-opus-5) re-derived the entropy, permanent, Gaussian, cycle and determinant calculations, checked the literature, and ran the computations summarized above on 2026-09-17; its verdict was MINOR GAPS. Its report is reproduced verbatim in [Appendix B](#). (3) No rewrite of this target was started by claude-fable-5-1 before its session ended; only the original writeup and referee report were present. (4) On 2026-09-18 Codex wrote the present note and moved the full proofs to [Appendix A](#). It repaired and marked four referee gaps: a finite-dimensional entropy-continuity proof for rounded fibre sizes; a uniform Fourier argument for the local lattice estimate, with [7]; a full factorial-moment enumeration explaining periodic projected walks; and an explicit invocation of the small-subgraph conditioning theorem from [8], with every hypothesis matched to a preceding formula. Citations for Bregman’s theorem and Radhakrishnan’s entropy proof were also added. No other mathematical content was changed. **No human mathematician has checked the argument.**

A Full proofs

Lemma 2.1 (Entropy inequality). *There is $c > 0$ such that every nonnegative 3×3 matrix with row sums $1/3$ satisfies*

$$F(\rho) \leq F(U) - c\|\rho - U\|_F^2, \quad F(U) = 2 \log(4/3).$$

Proof. Put $g(x) = 2x \log(1+x) - x \log x$ for $0 \leq x \leq 1$, with $0 \log 0 = 0$. We first prove

$$g(x_1) + g(x_2) + g(x_3) \leq 3g(1/3) \quad (x_1 + x_2 + x_3 = 1), \quad (5)$$

with equality only at the uniform point. Direct differentiation gives

$$g''(x) = \frac{x^2 + 2x - 1}{x(1+x)^2}.$$

Thus g is strictly concave below $r = \sqrt{2} - 1$ and strictly convex above it. A maximizer is interior because $g'(0+) = +\infty$. At an interior maximizer, two coordinates cannot both exceed r , since varying those two with fixed sum has positive second derivative.

If every coordinate is at most r , Jensen’s inequality proves (5). Otherwise, write the exceptional coordinate as $x > r$. Concavity on the other two gives

$$g(x_1) + g(x_2) + g(x_3) \leq \phi(x) := g(x) + 2g((1-x)/2).$$

For $1/3 \leq x < 1$,

$$\phi''(x) = \frac{x^4 - 10x^3 - 32x^2 + 34x - 9}{x(1+x)^2(1-x)(3-x)^2} < 0.$$

Indeed, the numerator is at most $-(103/3)x^2 + 34x - 9$, whose discriminant is -80 . Since $\phi'(1/3) = 0$, strict concavity gives $\phi(x) < \phi(1/3)$ for $x > 1/3$.

Apply (5) to each row of $x_{ab} = 3\rho(a, b)$. Since

$$F(\rho) = -\log 3 + \frac{1}{3} \sum_{a,b} g(3\rho(a, b)),$$

U is the unique maximizer. The Hessian of F at U is $-(9/8)I$ in the nine entry coordinates. Taylor’s theorem gives a strict quadratic decrease near U ; on the compact complement of a small neighbourhood, uniqueness gives a fixed gap, which may be weakened to the required quadratic bound. $\square$

Lemma 2.2 (Finite-type matching asymptotic). *Let A be a symmetric zero-one matrix of order s with constant row sum r , put $P = A/r$ and $u = (1/s, \dots, 1/s)$. Assume that $I - P^2$ is positive definite on $\mathbf{1}^\perp$ and that the vectors $e_j - e_k$ for columns occurring together in a row support generate the zero-sum integer lattice. If $Q_A(n; \alpha, \beta)$ is the probability that a uniform matching with integer type counts $n\alpha, n\beta$ uses only allowed type pairs, then, uniformly for $\|\alpha - u\| + \|\beta - u\| = O(n^{-2/5})$,*

$$Q_A(n; \alpha, \beta) = \left(\frac{r}{s}\right)^n \det_{\mathbf{1}^\perp}(I - P^2)^{-1/2} \times \exp\left\{\frac{ns}{2} [\|\beta - u\|^2 - \langle w, (I - P^2)^{-1}w \rangle]\right\} (1 + o(1)),$$

$$w = \beta - u - P(\alpha - u).$$

Proof. For every left vertex, independently choose one of its r permitted right types uniformly, and let S be the resulting vector of right-type counts. Counting matchings with a fixed type table in two ways gives the exact identity

$$Q_A(n; \alpha, \beta) = \left(\frac{r}{s}\right)^n \frac{\Pr(S = n\beta)}{\Pr(\text{Mult}(n, u) = n\beta)}. \quad (6)$$

The mean of S/n is $P\alpha$. At $\alpha = u$, its covariance divided by n , on $\mathbf{1}^\perp$, is

$$\Sigma_0 = \frac{1}{s}(I - P^2).$$

The normalized logarithmic moment generating function is

$$K_\alpha(t) = \sum_i \alpha_i \log \left(\sum_j P_{ij} e^{t_j} \right).$$

Uniformly near $(u, 0)$ on $\mathbf{1}^\perp$,

$$K_\alpha(t) = \langle P\alpha, t \rangle + \frac{1}{2} \langle t, \Sigma_0 t \rangle + cO(\|\alpha - u\| \|t\|^2 + \|t\|^3).$$

Writing $w = \beta - u - P(\alpha - u)$, the tilt with mean β is $t = \Sigma_0^{-1}w + cO((\|\alpha - u\| + \|\beta - u\|)^2)$, and its rate function is

$$I_\alpha(\beta) = \frac{1}{2} \langle w, \Sigma_0^{-1}w \rangle + \mathcal{O}((\|\alpha - u\| + \|\beta - u\|)^3). \quad (7)$$

Repair. The original writeup asserted uniform Fourier inversion. Here are the uniformity checks. After tilting, the type proportions remain in a fixed compact neighbourhood of u and the covariance on $\mathbf{1}^\perp$ remains uniformly positive definite. The characteristic function is a product of finitely many categorical characteristic functions. The lattice-generation hypothesis says that equality in the triangle inequality on the dual torus occurs only at the trivial character; compactness therefore gives modulus at most e^{-cn} outside a fixed neighbourhood of zero, uniformly in α, β . Inside that neighbourhood, the logarithm has the uniform expansion $-n \langle t, \Sigma t \rangle / 2 + cO(n \|t\|^3)$. Splitting at $\|t\| = n^{-2/5}$ makes the cubic error $o(1)$ on the Gaussian scale, the intermediate annulus is bounded by $e^{-cn^{1/5}}$, and rescaling t by $n^{-1/2}$ gives the Gaussian integral uniformly. This is the fixed-dimensional lattice Laplace argument of Greenhill–Janson–Ruciński [7]; all its nondegeneracy and lattice hypotheses are exactly the two assumptions of this lemma.

Consequently, the local probability in the numerator of (6) has the Gaussian prefactor on the full zero-sum lattice. The denominator has the same lattice covolume, so it cancels. The determinant ratio is $\det_{\mathbf{1}^\perp}(I - P^2)^{-1/2}$. Finally,

$$D(\beta \| u) = \frac{s}{2} \|\beta - u\|^2 + cO(\|\beta - u\|^3).$$

The deviations are $O(n^{-2/5})$, so multiplying the cubic errors by n gives $O(n^{-1/5}) = o(1)$. Substitution in (6), using $\Sigma_0^{-1} = s(I - P^2)^{-1}$, proves the displayed formula uniformly. $\square$

Proposition 3.1 (Second moment). *For the variable Z_n above,*

$$\frac{\mathbb{E}Z_n^2}{(\mathbb{E}Z_n)^2} \rightarrow R := 25 \left(\frac{16}{15}\right)^{20} \left(\frac{15}{23}\right)^8.$$

Moreover $\mathbb{E}Z_n = \Theta(n^{-5})(4/3)^{5n}$.

Proof. We first obtain a global bound. Bregman's theorem states that a zero-one matrix with positive row sums r_v satisfies

$$\text{per}(A) \leq \prod_v (r_v!)^{1/r_v}. \quad (8)$$

For the matching counted by $q_n(\rho, \sigma)$, a left vertex of type (a, b) has

$$n(1 - p_a - p_b + \sigma(a, b))$$

permitted partners, uniformly at least $n/4$ for large n . Applying (8), Stirling's formula, and then interchanging ρ, σ gives

$$\begin{aligned} \log q_n(\rho, \sigma) &\leq n \sum_{a,b} \rho(a, b) \log\left(\frac{1}{3} + \sigma(a, b)\right) + cO(\log n), \\ \log q_n(\rho, \sigma) &\leq \frac{n}{2}(L(\rho) + L(\sigma)) + \mathcal{O}(\log n). \end{aligned} \quad (9)$$

For the second line, use monotonicity of the logarithm to obtain

$$\sum_{a,b} (\rho(a, b) - \sigma(a, b)) [\log(\frac{1}{3} + \rho(a, b)) - \log(\frac{1}{3} + \sigma(a, b))] \geq 0.$$

Repair. For the rounded margins, rescale row a of ρ by $1/(3p_a)$, obtaining $\hat{\rho}$ with row sums $1/3$ and $\|\rho - \hat{\rho}\|_1 = O(1/n)$. In fixed dimension, the elementary entropy continuity bound $|H(x) - H(y)| \leq h_2(\delta) + \delta \log(8) = O(\delta \log(1/\delta))$ for $\delta = \|x - y\|_1/2$ gives an $O((\log n)/n)$ change in H . The derivative of $x \mapsto x \log(1/3 + x)$ is bounded on $[0, 1]$, so L changes by $O(1/n)$. Since $\|\hat{\rho} - U\|_F^2 \geq \frac{1}{2}\|\rho - U\|_F^2 - O(n^{-2})$, [Lemma 2.1](#) yields, uniformly,

$$F(\rho) \leq F(U) - c'\|\rho - U\|_F^2 + O((\log n)/n). \quad (10)$$

Stirling's formula in (2) gives

$$\log \Pr(\rho_i = \rho) = n(H(\rho) - 2H(p)) + O(\log n).$$

Also $H(p) = \log 3 + O(n^{-2})$. Combining this with (9), the first-moment asymptotic derived below, and the fact that every fibre of K_5 has degree 4, the contribution of a fixed five-overlap profile to (3) is at most

$$\exp \left\{ n \sum_{i=1}^5 (F(\rho_i) - F(U)) + O(\log n) \right\}. \quad (11)$$

Put $C_n = pp^T$. Since $C_n - U = O(1/n)$, (10)–(11) show that profiles with some $\|\rho_i - C_n\|_F > n^{-2/5}$ contribute at most a polynomial times $e^{-cn^{1/5}}$. There are only polynomially many profiles, so their total is $o(1)$.

It remains to evaluate the central window. Apply [Lemma 2.2](#) first to $A_0 = J_3 - I_3$. The eigenvalues of $P_0 = A_0/2$ are $1, -1/2, -1/2$, and the lattice hypothesis holds, whence

$$a_n = \frac{4}{3}(2/3)^n(1 + o(1)). \quad (12)$$

Together with Stirling's formula for M_n , (1) gives $\mathbb{E}Z_n = \Theta(n^{-5})(4/3)^{5n}$.

Let

$$\mathcal{T} = \{X \in \mathbb{R}^{3 \times 3} : X\mathbf{1} = X^T\mathbf{1} = 0\}, \quad z_i = 3\sqrt{n}(\rho_i - C_n).$$

This is four-dimensional. Uniformly in the central window, (2) and Stirling give

$$\Pr(\rho_i = \rho) = \frac{729}{(2\pi n)^2} e^{-\|z_i\|_F^2/2} (1 + o(1)). \quad (13)$$

The row-and-column-zero integer lattice has covolume 9: the basis $(e_a - e_3)(e_b - e_3)^T$, $a, b \in \{1, 2\}$, has Gram determinant 81. Hence the z_i -lattice has covolume $729/n^2$, matching (13).

For $A_1 = A_0 \otimes A_0$, $P_1 = A_1/4 = P_0 \otimes P_0$ has eigenvalues 1, $-1/2$ with multiplicity 4, and $1/4$ with multiplicity 4; on $\mathcal{T}$ it acts as multiplication by $1/4$. The lattice hypothesis again holds. Applying Lemma 2.2 and dividing by (12) squared yields

$$\frac{q_n(\rho_i, \rho_j)}{a_n^2} = \left(\frac{16}{15}\right)^2 \exp \left\{ \frac{4}{15} \langle z_i, z_j \rangle - \frac{1}{30} (\|z_i\|^2 + \|z_j\|^2) \right\} (1 + o(1)). \quad (14)$$

Let A now be the adjacency matrix of K_5 and set

$$M = \frac{19I_5 - 4A}{15}.$$

Equations (13)–(14) give this quadratic form independently in each of the four coordinates of $\mathcal{T}$. Its eigenvalues are $1/5$ once and $23/15$ four times, so it is positive definite. The central lattice sum therefore converges to its Gaussian integral, and

$$\frac{\mathbb{E} Z_n^2}{(\mathbb{E} Z_n)^2} \rightarrow \left(\frac{16}{15}\right)^{20} \det(M)^{-2} = 25 \left(\frac{16}{15}\right)^{20} \left(\frac{15}{23}\right)^8.$$

□

Proposition 3.2 (Short-cycle moments and variance identity). *For every fixed L , $(X_{3,n}, \dots, X_{L,n})$ converges jointly to independent Poisson variables of means $(\mu_3, \dots, \mu_L)$. For every finitely supported sequence of nonnegative integers b_ℓ ,*

$$\frac{\mathbb{E} [Z_n \prod_{\ell \geq 3} (X_{\ell,n})^{b_\ell}]}{\mathbb{E} Z_n} \rightarrow \prod_{\ell \geq 3} (\mu_\ell (1 + \delta_\ell))^{b_\ell}.$$

Finally,

$$\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 < \infty, \quad R = \exp \left(\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 \right).$$

Proof. A simple cycle in the lift projects to a cyclically nonbacktracking closed walk in the base. For a fixed rooted oriented projected walk of length ℓ , choosing distinct lift vertices gives $n^\ell(1 + O(1/n))$ possibilities, and the required matching edges occur with probability $n^{-\ell}(1 + O(1/n))$. The factorial-moment expansion for a fixed finite collection of lengths is dominated by vertex-disjoint cycles. Such choices factor. If two distinct cycles overlap, their union has strictly more edges than vertices and contributes $O(n^{v-e}) = O(1/n)$. Dividing by the 2ℓ roots and orientations of each lifted cycle gives the independent Poisson means $\mu_\ell = \text{tr}(B^\ell)/(2\ell)$.

Repair. This rooted count also handles a projected walk that is a proper power. If a primitive base walk is traversed d times, closure in the lift asks for a d -cycle of its monodromy permutation, whose expected number is $1/d$. Equivalently, its periodic word has only $2\ell/d$ distinct base root-orientations, and the factor $1/d$ restores the same normalization by 2ℓ . Thus periodic projections, such as doubled triangles at $\ell = 6$, are already included in $\text{tr}(B^\ell)/(2\ell)$; configurations with repeated lift vertices belong to the $O(1/n)$ degenerate terms.

Now bias the lift law by $Z_n/\mathbb{E}Z_n$. Equivalently, choose a uniform balanced colouring in each fibre and then independent uniform colour-respecting matchings. For r prescribed disjoint compatible edges in one matching, the probability that they all occur is

$$(3/(2n))^r(1 + o(1)), \quad (15)$$

uniformly for fixed r . This follows by deleting the endpoints and applying [Lemma 2.2](#) to the remaining counts, which differ from uniform by $O(1/n)$. The planted colours on a fixed vertex set are asymptotically independent and uniform. An ℓ -cycle has $2^\ell + 2(-1)^\ell$ proper 3-colourings, so its relative weight under the biased law is

$$3^{-\ell}(2^\ell + 2(-1)^\ell)(3/2)^\ell = 1 + \delta_\ell. \quad (16)$$

For the mixed factorial moment in the proposition, expand each falling factorial as an ordered selection of distinct cycles. Vertex-disjoint selections contribute the product of their rooted-walk counts and the factors (16), giving $\prod_\ell (\mu_\ell(1 + \delta_\ell))^{b_\ell}$. Every selection with an intersection has a union with $e > v$ and is $O(n^{v-e}) = O(1/n)$ after the same colour-compatible matching calculation. Since the total number and lengths are fixed, summing these errors is $o(1)$. This proves the asserted joint factorial moments.

It remains to identify the variance. Since the spectral radius of B is $3 < 4$ and $\text{tr } B = \text{tr } B^2 = 0$,

$$\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 = 2 \sum_{\ell \geq 1} \frac{\text{tr}(B^\ell)}{\ell 4^\ell} = -2 \log \det(I - B/4). \quad (17)$$

For completeness, the Ihara–Bass identity here is

$$\det(I - uB) = (1 - u^2)^5 \det(I - uA + 3u^2I). \quad (18)$$

It follows by writing $B = XY - J$, where J reverses directed edges and X, Y are the head and tail incidence matrices, then using $J^2 = I$, $YX = A$, $YJX = 4I$ and the determinant lemma. At $u = 1/4$,

$$I - A/4 + 3I/16 = (15/16)M,$$

so $\det(I - B/4) = (15/16)^{10} \det M$. Equations (17) and [Proposition 3.1](#) now give

$$\exp \left(\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 \right) = (16/15)^{20} \det(M)^{-2} = R.$$

□

Theorem 1.2 (Chromatic number of a random lift of K_5). *Let H_n be a uniform random n -lift of K_5 . Then*

$$\Pr(\chi(H_n) = 3) \longrightarrow 1.$$

Proof. We apply the small-subgraph conditioning theorem in the form stated in [8, Section 4] to $Y_n = Z_n/\mathbb{E}Z_n$ and the cycle counts $X_{\ell,n}$.

Repair. The theorem's hypotheses are all explicit above: (i) finite collections of $X_{\ell,n}$ converge to independent Poisson variables with means μ_ℓ by [Proposition 3.2](#); (ii) their mixed factorial moments under the Z_n -size-biased law converge with factors $1 + \delta_\ell$, again by [Proposition 3.2](#); (iii) $\delta_\ell > -1$ and $\sum \mu_\ell \delta_\ell^2 < \infty$; and (iv) the normalized second moment tends to $\exp(\sum \mu_\ell \delta_\ell^2)$ by [Propositions 3.1](#) and [3.2](#). Therefore the small-subgraph conditioning theorem applies directly, including the required exchange of the limits in the number of conditioned cycle lengths and in n .

It yields

$$Y_n \xrightarrow{d} W := \prod_{\ell \geq 3} (1 + \delta_\ell)^{P_\ell} e^{-\mu_\ell \delta_\ell}, \quad (19)$$

where the P_ℓ are independent Poisson variables of means μ_ℓ , and it gives $\Pr(Z_n > 0) \rightarrow 1$. To see positivity directly, the logarithm of the partial products is

$$\sum_{\ell=3}^L (P_\ell - \mu_\ell) \log(1 + \delta_\ell) + \sum_{\ell=3}^L \mu_\ell (\log(1 + \delta_\ell) - \delta_\ell).$$

The first series converges almost surely because its variances are summable, and the second converges absolutely; hence $W > 0$ almost surely. Thus H_n is asymptotically almost surely 3-colourable. Notice that the rounded $t_a = n/3 + O(1)$ were used throughout, so this holds for every integer sequence n , not only multiples of 3.

We finally exclude chromatic number 2. In a proper 2-colouring let s_i be the number of colour-1 vertices in fibre i . Every base-edge matching forces $s_i + s_j = n$. Since K_5 contains triangles, $s_i = n/2$ for all i ; hence no such colouring exists for odd n . For even n , the expected number of proper 2-colourings is exactly

$$\binom{n}{n/2}^5 \left(\frac{((n/2)!)^2}{n!} \right)^{10} = \binom{n}{n/2}^{-5} = o(1).$$

Markov’s inequality shows that H_n is asymptotically almost surely nonbipartite. Together with 3-colourability, this proves $\Pr(\chi(H_n) = 3) \rightarrow 1$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/chromatic_number_of_random_lifts_of_complete_graphs/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The OPG entry (attributed to Amit, Linial, Matoušek [ALM02]) asks: *is the chromatic number of a random lift of K_5 concentrated on a single value?* The discussion fixes the model (uniform random h -lift = independent uniform perfect matching per base edge, $h \rightarrow \infty$) and records that the value is 3 or 4.

The writeup proves $\Pr(\chi(H_n) = 3) \rightarrow 1$ for H_n a random n -lift of K_5 , using exactly that model (Section 1: “for each $uv \in E$, a uniform perfect matching between the fibres”, independent across the 10 edges of K_5). “Asymptotically almost surely” is in $n = h$, the lift order, with the base K_5 fixed — the intended asymptotics. The writeup separately excludes $\chi = 2$ (Section 8), so it really delivers a single value, not just an upper bound. This is the intended question, answered affirmatively with the value identified; a reasonable author of the original problem would consider the K_5 question resolved.

Two scope points, neither an interpretation game:

- The OPG entry also asks the general question for K_n . The writeup does not address it (its “Scope” section only extends the *upper* bound $\chi \leq 3$ to arbitrary fixed simple 4-regular bases). The verdict block does not claim otherwise.
- The writeup works with **fibre-balanced** (“strongly equitable”) colourings, i.e. each fibre gets $t_a \in \{\lfloor n/3 \rfloor, \lceil n/3 \rceil\}$ vertices of colour a . That is a legitimate restriction: $Z_n > 0$ implies 3-colourability. It is the same object NPG call Y .

B.2 Step-by-step findings

I re-derived every formula independently; “re-derived” below means I obtained the same expression from scratch, not that I read the writeup’s derivation.

step	label	note
1. (1) $E Z_n = M_n^5 a_n^{10}$	VALID	Matchings independent across the 10 base edges; a_n depends only on the fibre colour counts, which are the same t for every fibre, so it is the same for all 10 edges and all fibre-balanced colourings.
2. (2) $\Pr(p_i = p) = \Pi_a(t_a!)^2 / (n! \Pi_{ab}(np_{ab})!)$	VALID	Re-derived: $\#\{\text{colourings } 2 \text{ with overlap } p\} = \Pi_a t_a! / \Pi_b (np_{ab})!$, divided by $n! / \Pi t_a!$.

step	label	note
3. (3) $E Z_n^2 / (E Z_n)^2 = E_{\rho} \prod_{ij \in E} q_n(\rho_i, \rho_j) / a_n^2$	VALID	Conditional on the two colourings, the 10 matchings are independent, and the pair-properness probability for edge ij depends only on (ρ_i, ρ_j) .
4. Reduction $F(\rho) = -\log 3 + (1/3) \sum g(3\rho_{ab})$, $g(x) = 2x \log(1+x) - x \log x$	VALID	Re-derived exactly: $H(\rho) = -(1/3) \sum x \log x + \log 3$, $L(\rho) = (1/3) \sum x \log(1+x) - \log 3$ with $x = 3\rho$. $F(U) = 2 \log(4/3)$ confirmed.
5. $g'(x) = (x^2 + 2x - 1)/(x(1+x)^2)$, sign change at $\sqrt{2}-1$	VALID	Re-derived; matches numerics to $1e-6$ (script 01).
6. Lemma 1, case “all coordinates $\leq r$ ”: Jensen	VALID	g concave on $[0, r]$, $r = \sqrt{2}-1 > 1/3$, so $\sum g(x_i) \leq 3g(1/3)$.
7. Lemma 1, “no interior maximizer has two coordinates $> r$ ”	VALID	Moving two convex-region coordinates with fixed sum has positive second derivative, so such a point is a local minimum in that direction. $g'(0+) = +\infty$ rules out boundary maximizers.
8. Lemma 1, one-coordinate case: $\phi(x) = g(x) + 2g((1-x)/2)$	VALID	The other two coordinates are $\leq r$ and their mean $(1-x)/2 \leq (1-r)/2 < r$, so concavity applies.
9. ϕ'' numerator $N(x) = x^4 - 10x^3 - 32x^2 + 34x - 9$	VALID	I expanded $(x^2+2x-1)(1-x)(3-x)^2 + (x^2-6x+1)x(1+x)^2$ by hand and got exactly this polynomial.
10. $N(x) \leq -(103/3)x^2 + 34x - 9$ on $[1/3, 1]$, discriminant -80	VALID	The bound is equivalent to $x^2 - 10x + 7/3 \leq 0$, true for $x \geq 0.2390$; $34^2 - 4 \cdot (103/3) \cdot 9 = 1156 - 1236 = -80$. Numerically $\max N$ on $[1/3, 1] = -1.0488$.
11. Lemma 1 conclusion + Hessian $-(9/8)I$ at U , quadratic gap (4)	VALID	$3g'(1/3) = -9/8$ confirmed. Compactness + uniqueness $\Rightarrow$ (4). Numerically the max of $\sum g$ over the simplex is attained only at $(1/3, 1/3, 1/3)$ for $d = 4$ (script 01c).
12. (6) uniform version for rounded row sums p	GAP (routine)	The continuity-of-entropy estimate (“changes the matrix by $O(1/n)$ in total variation, entropy by $O(\log n / n)$ ”) is asserted, not proved; it is the standard Fannes-type bound and the $O(\log n)$ it produces is swamped by $\exp(-c n^{1/5})$.
13. (7) Bregman/Minc: $\text{per}(A) \leq \prod_v (r_v!)^{1/r_v}$	VALID	Correctly stated theorem (Bregman 1973; entropy proof Radhakrishnan 1997); the writeup’s sketch is Radhakrishnan’s argument.
14. Row sums $r_v = n(1 - p_a - p_b + \sigma(a, b)) \geq n/4$	VALID	Inclusion-exclusion over the two colour classes; $1 - 2/3 + \sigma \geq 1/3 - O(1/n)$.
15. (8) $\log q_n(\rho, \sigma) \leq n \sum \rho_{ab} \log(1/3 + \sigma_{ab}) + O(\log n)$	VALID	Re-derived: $\log(r!)/r = \log r - 1 + O(\log r / r)$, subtract $\log n! = n \log n - n + O(\log n)$; the per-row error sums to $O(\log n)$ since $r_v \geq n/4$. Verified numerically against the exact rate (script 05: max violation $-1.4e-5$ over 19851 random pairs).
16. (9) symmetrisation	VALID	$\sum (\rho - \sigma)(\log(1/3 + \rho) - \log(1/3 + \sigma)) \geq 0$ termwise since $\log$ is increasing.
17. (10) $\log q_n \leq (n/2)(L(\rho) + L(\sigma)) + O(\log n)$	VALID	Average of (8) and its transpose ($\text{per } A = \text{per } A^T$). Numerically confirmed: $\max \lambda(a, b) - (L(a) + L(b))/2 = -1.3e-3$ over 19851 pairs.
18. Lemma 2 identity (12): $Q = (r/s)^n \Pr(S=n\beta) / \Pr(\text{Mult}(n, u)=n\beta)$	VALID	Re-derived exactly from $\text{per}(M) = \prod (n\alpha_i)! \prod (n\beta_j)! \sum_m 1 / \prod m_{ij}!$.
19. Lemma 2 covariance $\Sigma_0 = (1/s)(I - P^2)$ on $1\perp$	VALID	$\sum_i u_i [\text{diag}(P_{\cdot i}) - P_{\cdot i} \{P_{\cdot i}\}^T] = (1/s)(I - P^2)$ using double stochasticity and symmetry of P .

step	label	note
20. Lemma 2 local CLT / Fourier inversion / lattice constant	GAP	This is the one genuinely analytic step and it is a sketch ("Fourier inversion under this tilt gives a Gaussian local prefactor"). The lattice hypothesis is verified correctly (for A_0 and $A_1 = A_0 \otimes A_0$ the only maximum-modulus character is trivial), the tilting expansion and the $O(\delta^3)$ rate-function error are right, and $n \cdot O(n^{-6/5}) = o(1)$ as claimed; but uniformity of the $(1+o(1))$ is asserted. This is exactly the Greenhill–Janson–Ruciński-type lemma NPG invoke by citation over several pages. Repairable by citation. The formula itself passes a nontrivial internal consistency test: its exponent is symmetric under $(\alpha, \beta) \leftrightarrow (\beta, \alpha)$, as $\text{per}(A) = \text{per}(A^T)$ demands.
21. (14) $a_n = (4/3)(2/3)^n(1+o(1))$	VALID	$\det_{\{1\}}(I - P_0^2)^{-1/2} = (3/4)^{-1} = 4/3$ (numerics). Exact a_n computed by inclusion–exclusion for $n \leq 30$: ratio $a_n / [(4/3)(2/3)^n] = 0.8438, 0.9492, \dots, 0.9888 \rightarrow 1$.
22. (15) $E Z_n = \Theta(n^{(-5)(4/3)})\{5n\}$	VALID	Exact values for $n \leq 30$ give $E Z_n / [n^{(-5)(4/3)}\{5n\}] = 4.72, 5.19, 5.49 \rightarrow$ the predicted constant $(3\sqrt{3})^5(4/3)^{10}/(2\pi)^5 = 6.87$.
23. (16) $\text{spec}(P_1) = \{1, -1/2 (\times 4), 1/4 (\times 4)\}$	VALID	Numerics.
24. (17) $\log \Pr(\rho_i = \rho) = n(H(\rho) - 2H(p)) + O(\log n)$	VALID	Re-derived by Stirling on (2).
25. (18) $\text{contribution} \leq \exp\{n \Sigma(F(\rho_i) - F(U))\}$	VALID	Each fibre lies in 4 base edges, so $\Sigma_{\{ij \in E\}}(1/2)(L(\rho_i) + L(\rho_j)) = 2 \Sigma_i L(\rho_i)$; the normalisation $2 \log 3 + 4 \log(2/3)$ equals $F(U)$ exactly. This is where 4-regularity enters, as stated.
26. (19) non-central overlaps contribute $o(1)$	VALID	$n \cdot c' \cdot (n^{-2/5})^2 = c' \cdot n^{1/5}$; the number of profiles is $O(n^{45})$.
27. (20) $\Pr(\rho_i = \rho) = 729/(2\pi n)^2 \exp(-\ z\ ^2/2)(1+o(1))$; covolume $729/n^2$	VALID	Gram matrix of the basis $(e_{a-e_3})(e_{b-e_3})^T$ is $G \otimes G$ with $G = [[2, 1], [1, 2]]$, $\det = 3^2 \cdot 3^2 = 81$, covolume 9; the z -lattice then has covolume $(3/\sqrt{n})^4 \cdot 9 = 729/n^2$. Independent cross-check: a standard Gaussian on T has entrywise variance $[(I-J)/3]_{aa} = 4/9$, and the exact hypergeometric variance of $n\rho_{ab}$ is $4n/81$, i.e. $\text{Var}(3\sqrt{n} \rho_{ab}) = 4/9$. Consistent. Also $E \rho_i = C_n$ exactly.
28. (21) $q_{n/a_n} = (16/15)^2 \exp\{(4/15)\langle z_i, z_j \rangle - (1/30)(\ z_i\ ^2 + \ z_j\ ^2)\}(1+o(1))$	VALID	Re-derived from (11) with $s=9, r=4, P_1 = 1/4$ on T : prefactor $(9/16) \cdot \det_{\{1\}}(I - P_1^2)^{-1/2} = (9/16)(64/45)^2 = (16/15)^2$ (numerics agree to $1e-15$); the exponent reduces to exactly the stated form.
29. (22)–(23) $M = (19I_5 - 4A)/15$, $\text{spec} \{1/5, 23/15 (\times 4)\}$	VALID	Numerics: $\text{eig}(M) = (0.2, 1.5333 \times 4)$; positive definite.
30. (24)–(25) $R = (16/15)^{20} \det(M)^{-2} = 25(16/15)^{20(15/23)} 8 = 2.974554$	VALID	$\det M = 1.1055447 = (1/5)(23/15)^4$; $\det(M \otimes I_4)^{-1/2} = \det(M)^{-2}$.
31. (26) $\mu_\ell = \text{tr}(B^\ell)/(2\ell)$, $\delta_\ell = 2(-1/2)^\ell$, Poisson limits	VALID	$\text{tr}(B^3)=60, \text{tr}(B^4)=120, \text{tr}(B^5)=120 \Rightarrow \mu_3=10, \mu_4=15, \mu_5=12$, matching the 10 triangles and 15 four-cycles of K_5 (each base cycle contributes a Poisson(1) fixed-point count). Simulation over 1400 random lifts: mean triangles 9.79–10.20, 4-cycles 14.75–15.15, 5-cycles 11.72–12.13.
32. Size-biased construction (uniform balanced colouring per fibre, then colour-respecting matchings)	VALID	Correct because the properness probability a_n^{10} is the same for every fibre-balanced colouring.
33. (27) $\Pr(r \text{ prescribed disjoint compatible edges}) = (3/2n)^r(1+o(1))$	VALID	A colour- a vertex has $\approx 2n/3$ admissible partners.
34. (28) $3^{(-\ell/2)} \ell + 2(-1)^{\ell(3/2)} \ell = 1 + \delta_\ell$	VALID	$P(C_\ell, 3) = 2^\ell \ell + 2(-1)^\ell \ell$; the planted colours on a fixed vertex set are asymptotically i.i.d. uniform.
35. (29) joint factorial moments	GAP (routine)	The disjoint-cycle/overlapping-union enumeration is only sketched, as in the unbiased case. Standard.

step	label	note
36. (30) $\sum_{\ell \geq 3} \mu_\ell \delta_\ell^2 = -2 \log \det(I - B/4)$	VALID	$\text{tr } B = \text{tr } B^2 = 0$ (a length-2 closed non-backtracking walk would need $(u,v) \rightarrow (v,u)$); spectral radius of B is $3 < 4$. Exactly the Ihara–Bass formula with
37. (31) Ihara–Bass $\det(I - uB) = (1 - u^2)^5 \det(I - uA + 3u^2I)$	VALID	
38. (32) $R = \exp(\sum \mu_\ell \delta_\ell^2)$	VALID	$I - A/4 + (3/16)I = (15/16)M$ exactly (max entrywise difference 0). Numerically $\sum_{\ell=3}^{60} \mu_\ell \delta_\ell^2 = 1.090094126$ vs $\log R = 1.090094129$ (difference $3e-9$, pure truncation). This exact match is a strong global consistency check on Sections 3–6. $E[(1+\delta)^\mu P] = e^{\mu\delta}$ for $P \sim \text{Poisson}(\mu)$.
39. §7 W_L martingale, $E W_L^2 = \exp(\sum_{\ell=3}^L \mu_\ell \delta_\ell^2)$	VALID	
40. §7 $W > 0$ a.s.	VALID	$\delta_\ell \in [-1/4, 1/4]$ for $\ell \geq 3$, so $\log(1+\delta_\ell)$ is finite; variances summable; second series absolutely convergent.
41. §7 conditioning $\Rightarrow Y_n \rightarrow W \Rightarrow \Pr(Z_n = 0) \rightarrow 0$	GAP	This is Janson’s small-subgraph-conditioning theorem. The writeup “includes the conditioning argument” but the step “at each fixed count vector the limiting conditional mean is the corresponding likelihood ratio W_L ” is asserted; joint convergence of $(Y_n, X_{\{3,n\}}, \dots, X_{\{L,n\}})$ is what actually needs proof, and the interchange of the $L \rightarrow \infty$ and $n \rightarrow \infty$ limits is glossed. All four hypotheses of Janson (1995) are, however, verified in the writeup, so this is repairable by a precise citation.
42. §8 $s_i + s_j = n$ for every base edge, hence n even and $s_i = n/2$	VALID	A matching sends colour-1 vertices of fibre i bijectively onto colour-2 vertices of fibre j ; K_5 contains triangles.
43. §8 $E[\#2\text{-colourings}] = C(n, n/2)^5 \cdot C(n, n/2)^{-10} = o(1)$	VALID	Probability that one matching respects a balanced 2-colouring is $((n/2)!^2/n!) = 1/C(n, n/2)$. Simulation: 0 bipartite lifts in 1400+ samples.
44. “Scope”: extension to any fixed simple 4-regular base (upper bound only)	VALID	The generalisation is consistent: for a $d=4$ base with

B.3 Reference check

The writeup is nearly self-contained: it names no paper. The external results it uses are:

1. **Bregman–Minc, $\text{per}(A) \leq \prod_v (r_v!)^{1/r_v}$ for 0-1 matrices, (7).** Confirmed: conjectured by Minc (1963), proved by Bregman (1973); entropy proof by J. Radhakrishnan, “An entropy proof of Bregman’s theorem”, JCTA 77 (1997) 161–164. The writeup’s sketch (expose rows in random order; the rank of a row among the r_v rows meeting its support is uniform, giving expected $\log(r_v!)/r_v$) is Radhakrishnan’s proof. Statement and hypotheses used correctly.
2. **Ihara–Bass determinant identity, (31).** Proved inline; verified numerically.
3. **Local CLT / Laplace summation over lattices, Stirling, Janson’s small subgraph conditioning.** Not cited; sketched inline (see GAPs above).

Literature status — the decisive check. The relevant modern reference is J. D. Nir, X. Pérez-Giménez, *The chromatic number of random lifts of complete graphs*, arXiv:2109.13347 (I retrieved and read the full PDF text). Their thresholds:

- Theorem 1.1: $d \geq u_k := 2 \log k / (\log k - \log(k-1)) \Rightarrow$ a.s. **not** k -colourable. $u_3 = 5.41902$.
- Theorem 1.2: $d < \ell_k := 2(k-1)^3 \log(k-1) / (k(k-2)) \Rightarrow$ a.s. k -colourable. $\ell_3 = \mathbf{3.69678}$.
- Corollary 1.3(iii): $d \in [\ell_k, u_k] \Rightarrow$ only $\chi \in \{k, k+1\}$.

Since $d = 4$ lies in $[3.69678, 5.41902]$, **NPG explicitly leave the K_5 case open** — their introduction says so (“it is not known for $G = K_k$ with $k \geq 5$ ”). So the writeup’s theorem is *not* already known from NPG.

Why NPG stop at $d < \ell_3$ is exactly the point where the writeup improves on them. In their §5.2 they bound the second-moment exponent $f(A, B)$ (their (38)) by **relaxing the marginal constraints (35) on B to (39)**, obtaining $g(A) = \sum_v [h(A_v) + (d/2) \log(1 - 2/k + \rho(A_v))]$ — the *pairing-model* exponent — and then apply Achlioptas–Naor (their Theorem 3.1), which needs $d/2 < c_k$, i.e. $d < \ell_k$. I verified numerically that this relaxation genuinely fails at $d = 4$: along $a_{ii} = (1+2t)/9$, $a_{ij} = (1-t)/9$, g increases from 0.575364 ($t=0$) to 0.580944 ($t=0.4$), so $\tilde{A}$ is not the maximiser of the relaxed bound. The writeup instead keeps the matching constraint and bounds the permanent by Bregman, which is a *strictly sharper* relaxation that is tight at U — and for $d = 4$ its maximiser is U . Both facts are confirmed below. So there is no contradiction with NPG, and no priority problem.

No later paper resolving $d = 4$ was found (searches for 2022–2025 work on random lifts / chromatic number / K_5 return only NPG 2021 and Farzad–Theis arXiv:1003.1527). The earlier references [ALM02] and [FT12] exist as described in the OPG entry and are not used in the proof.

B.4 Computational check

Scripts in `verification_astra/scripts/chromatic_number_of_random_lifts_of_complete_graphs/` (python3, numpy/scipy):

- `01_entropy.py` — Lemma 1.
 - $g'(x)$ numeric vs formula agree to $1e-6$; $3g'(1/3) = -1.125 = -9/8$.
 - max of $N(x) = x^4 - 10x^3 - 32x^2 + 34x - 9$ on $[1/3, 1]$ is $-1.0488 < 0$; max $\varphi' = -0.3099$; max $\{x \geq 1/3\} \varphi(x) = \varphi(1/3) = 1.6739764336$.
 - max of $g_d(x_1) + g_d(x_2) + g_d(x_3)$ on the simplex: for $d = 4$ it is **1.6739764336, attained only at $(1/3, 1/3, 1/3)$** — Lemma 1 holds. For $d = 3$ it also holds; for $d = 4.5, 5, 6$ the uniform point is *not* the maximum (e.g. $d = 5$: 1.86257 at (0.0916, 0.0916, 0.8167) vs 1.81782 at uniform). The local criterion $g_d'(1/3) < 0$ is $d < 96/21 = 4.5714$, so $d = 4$ is inside and $d = 5$ outside — i.e. the method is specific to the 4-regular case, as the writeup’s “this is exactly where 4-regularity is used” says.
 - Direct 9-variable maximisation of F over row-stochastic/3 matrices returns max $F = 0.5753641449 = F(U) = 2\log(4/3)$.
- `02b_exact_exponent_fast.py`, `05_bregman_check.py` — the *exact* second-moment exponent, bypassing the writeup’s bound entirely. From $\text{per}(A)/n!$ by Stirling, $\lambda(\rho, \sigma) := \lim (1/n) \log q_n(\rho, \sigma) = -h(\rho) - h(\sigma) + H(B)$, B the maximum-entropy coupling of (ρ, σ) supported on $\{(i, j, i', j') : i \neq i', j \neq j'\}$ (computed by Sinkhorn). This is exactly NPG’s $f(A, B)$ maximised over B .
 - $\lambda(U, U) = -0.810930216 = \log(4/9)$; $\Phi(U, \dots, U) = 2.87682072 = 2(5\log 3 + 10\log(2/3))$, i.e. the exponent of $(E Z_n)^2$. Encoding validated.
 - **(8) holds:** max over 19851 random doubly-stochastic pairs of $\lambda(a, b) - \sum a \log(1/3 + b) = -1.38e-5 (\leq 0)$.
 - **(10) holds:** max of $\lambda(a, b) - (L(a) + L(b))/2 = -1.35e-3 (\leq 0)$.
 - **The exact per-fibre exponent $h(\rho) + 2\lambda(\rho, \rho)$ is maximised at U :** global search returns $0.57536414 = 2\log(4/3)$ at U ; 20000 random restarts never exceed it (best 0.57525791 at a matrix within 0.006 of U). Along the family above it is strictly decreasing ($0.575364 \rightarrow 0.568501$ at $t = 0.2 \rightarrow 0.287682$ at $t = 1$), whereas NPG’s relaxed g increases. So the conclusion of Lemma 1 is true for the *exact* exponent too, not only for the writeup’s upper bound.
- `03_cycles_variance.py` — Sections 3–6.
 - $\text{spec}(P_0) = (-1/2, -1/2, 1)$; $\det_{\{1\}}(I - P_0^2)^{\wedge} \{-1/2\} = 1.3333 = 4/3$.
 - $(9/16) \cdot \det_{\{1\}}(I - P_1^2)^{\wedge} \{-1/2\} = 1.13777778 = (16/15)^2$.
 - $\text{eig}(M) = (0.2, 1.5333 \times 4)$; $\det M = 1.10554469$; **$R = 2.9745540512$** , $\log R = 1.0900941290$.
 - B is 20×20 with row sums 3; $\text{tr}(B^{\wedge} \ell) = 0, 0, 60, 120, 120, 780, 2520, 6120$ for $\ell = 1..8$; $\mu_3 = 10, \mu_4 = 15, \mu_5 = 12, \mu_6 = 65$.
 - **$\Sigma_{\{t=3\}}^{\wedge} \{60\} \mu_t \delta_t^2 = 1.090094126$ vs $\log R = 1.090094129$** — the writeup’s exact variance match (32) is confirmed (residual $3e-9$ is truncation). $-2 \log \det(I - B/4) = 1.0900941290$ to 10 digits.
 - Ihara–Bass verified at $u = 0.1, 0.25, 0.3$; $I - A/4 + (3/16)I - (15/16)M = 0$ exactly.
 - Exact $E Z_n$ for $n \leq 30$ (permanents by inclusion–exclusion): $a_n / [(4/3)(2/3)^{\wedge} n] = 0.8438, 0.9492, 0.9611, 0.9717, 0.9773, 0.9812, 0.9859, 0.9888 \rightarrow 1$, and $E Z_n / [n^{\wedge (4/3)} \{5n\}] = 0.43, 2.35, 3.19, 3.91, 4.37, 4.72, 5.19, 5.49$, converging to the predicted 6.87.
- `04_simulate_lifts.py` — random n -lifts of K_5 , exact 3-colourability by degree- <3 kernelisation + MRV/forward-checking backtracking:

n	vertices	trials	3-colourable	fraction	mean triangles	mean C_4	mean C_5	bipartite
3	15	400	211	0.527	9.85	15.14	11.84	0
5	25	400	342	0.855	9.86	14.75	11.72	0
8	40	300	275	0.917	10.20	15.14	12.13	0
12	60	300	293	0.977	9.97	15.15	12.07	0
20	100	200	200	1.000	9.79	14.94	12.09	0

The fraction rises monotonically to 1, consistent with a.a.s. 3-colourability; the cycle means match $\mu_3 = 10, \mu_4 = 15, \mu_5 = 12$; no bipartite lift ever occurred.

Nothing failed. Every explicit constant in the writeup $(4/3, (16/15)^2, 1/5$ and $23/15, R = 25(16/15)^{20(15/23)}8, \mu_3 = 10, \mu_4 = 15$, the (32) identity) reproduces exactly.

B.5 Caveats

1. **Two sketched analytic steps.** Lemma 2’s uniform local CLT (step 20) and the small-subgraph-conditioning limit (step 41) are the only places where the writeup asserts rather than proves. Both are standard: NPG spend most of their ~35 pages on the analogues (they cite Greenhill–Janson–Ruciński for the Laplace-summation/local limit step and Robinson–Wormald/Janson for the conditioning). A referee would require these to be written out or replaced by precise citations with hypotheses checked. I judge them repairable, not holes, because the writeup does verify all four Janson hypotheses explicitly and because Lemma 2’s formula passes the (α, β) -symmetry test and reproduces (32) exactly.
2. **The joint factorial-moment computation (29)** is a one-sentence “the same disjoint-cycle and overlapping-union enumeration”; routine but unwritten.
3. **(6)**, the passage from exact row sums $1/3$ to the rounded p , uses an unproved continuity-of-entropy estimate; harmless (the resulting $O(\log n)$ is dominated by $\exp(-c n^{\wedge (1/5)})$), but it is where the “all integers n , not merely multiples of 3” claim ultimately rests. NPG needed a separate section (their §7) for the non-divisible case; the writeup’s set-up is general from the start, so this is plausible, but it is the thinnest part of the “for all n ” claim.

4. **Non-primitive closed walks.** $\mu_\ell = \text{tr}(B^\ell)/(2\ell)$ counts closed non-backtracking walks including proper powers (e.g. a triangle traversed twice for $\ell = 6$), whose symmetry group is smaller than 2ℓ . The writeup’s “dividing rooted oriented cycles by 2ℓ ” does not address this. I checked $\ell = 6$ by hand: doubled triangles contribute 60 to $\text{tr}(B^6) = 780$, i.e. $5 = 10 \times \mathbb{E}[\#2\text{-cycles of a uniform permutation}]$ to $\mu_6 = 65$, which is exactly right — so the formula survives, but the justification as written is incomplete. Also, degenerate lifts (fixed points giving a non-simple walk) are swept into the “ $(1+O(1/n))$ ”.
5. **Scope of the claim.** The theorem is for K_5 only; the OPG entry’s more general question about K_n is untouched. The “Scope” section’s extension to arbitrary fixed simple 4-regular bases gives 3-colourability ($\chi \leq 3$) only, which the writeup states correctly — for a bipartite base $\chi = 2$, so “ $\chi = 3$ ” would be false there.
6. **The method is knife-edge in d.** Lemma 1 holds for $d \leq 4$ and fails for $d \geq 5$ (the local criterion is $d < 96/21 = 4.571$). So the companion open case $d = 5$ (lifts of K_6 , also in NPG’s gap $[\ell_3, u_3]$) is *not* resolved by this argument. The writeup does not claim it, but a reader might over-read the “Scope” paragraph.
7. **Edge cases.** n odd is handled (Section 8 forces n even for a 2-colouring, and the balanced-colouring set-up allows $t_a = \lceil n/3 \rceil$ or $\lfloor n/3 \rfloor$). Small n is irrelevant to an a.a.s. claim; my simulation shows 47% of 3-lifts are not 3-colourable, which is consistent. Disconnected lifts occur with probability $o(1)$ and change nothing.
8. **Novelty self-caveat.** The writeup’s own caveat (“Novelty relative to the full cited literature has not been independently checked”) is now discharged: I checked it, and the result is new relative to NPG 2021 and everything I could find.

B.6 Referee summary

I set out to break this writeup and could not. The suspicious step — a second-moment argument succeeding exactly where Nir-Pérez-Giménez (arXiv:2109.13347) stop — turns out to be a legitimate and identifiable improvement rather than an error: NPG bound the lift’s second-moment exponent by *dropping* the perfect-matching marginal constraints (their (35) $\rightarrow$ (39)), which turns the problem into the pairing-model/Achlioptas-Naor optimisation and forces $d < \ell_3 = 3.697$; the writeup instead keeps the matching and bounds the permanent by Bregman’s inequality, a relaxation that is tight at the uniform overlap and whose maximiser at $d = 4$ is the uniform overlap. I re-derived every formula in the writeup independently — (1), (2), (3), the reduction to g , the ϕ ’s numerator (by hand expansion), (8), (10), (12), the Σ_0 covariance, (17), (18), (20) (including the lattice covolume 729 and an independent hypergeometric-variance cross-check), (21) (including the $(16/15)^2$ prefactor and an (α, β) -symmetry consistency test), (22)–(25), (30), (31), (32), and Section 8 — and found no arithmetic or algebraic error. I also computed the *exact* second-moment exponent (max-entropy couplings, bypassing the writeup’s bound) and confirmed both that (8) and (10) hold over ~ 20000 random overlap pairs and that the exact exponent is globally maximised at the uniform overlap; the variance identity $\Sigma \mu_\ell \delta_\ell^2 = \log R = 1.0900941$ matches to $3e-9$; and simulated random lifts are 3-colourable in 100% of 200 trials at $n = 20$ with cycle counts matching $\mu_3 = 10$, $\mu_4 = 15$, $\mu_5 = 12$. What keeps this from CONFIRMED is not a suspected error but missing rigour in two standard technical steps the writeup chose to sketch rather than cite — the uniform local limit theorem of Lemma 2 and the passage from the four verified Janson hypotheses to $Z_n > 0$ a.a.s. in Section 7 — plus the incompletely justified treatment of non-primitive closed walks in μ_ℓ . Those are, in my judgement, routine and repairable, so: **MINOR_GAPS**, with the recommendation that the writeup be resubmitted with §3 and §7 either written out or reduced to precise citations (Greenhill-Janson-Ruciński; Janson 1995 / Wormald’s survey). If it survives that, it is a genuine resolution of the Amit-Linial-Matoušek K_5 question and is publishable.

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